

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287515

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

ADACHI; Koki et al.

ELECTRONIC CONTROL UNIT

Abstract

An electronic control unit includes a substrate on which electronic components are mounted; a plurality of terminals having first ends connected to the substrate; a plurality of electric wires having a first end portion connected to one of second ends of the plurality of terminals on an opposite side to the first ends; and a routing board having a base in a shape of a plate, arranged in parallel with the substrate. The routing board routes the plurality of electric wires introduced from outside on a routing surface of the base opposite to a back surface facing the substrate. The second ends of at least one of the terminals are located between the base and the substrate. The base has an opening partially facing the terminals having the second ends located between the base and the substrate and introducing the electric wires.

Inventors: ADACHI; Koki (Shizuoka, JP), HAMADA; Ryo (Shizuoka, JP)

Applicant: YAZAKI CORPORATION (Tokyo, JP)

Family ID: 96810598

Assignee: YAZAKI CORPORATION (Tokyo, JP)

Appl. No.: 19/069442

Filed: March 04, 2025

Foreign Application Priority Data

JP 2024-036555

Mar. 11, 2024

Publication Classification

Int. Cl.: H05K5/00 (20250101); B60R16/023 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application is based on, and claims priority from the Japanese Patent Application No. 2024-036555, filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an electronic control unit.

BACKGROUND

[0003] Conventionally, vehicles such as automobiles are equipped with an electronic control unit used for controlling various functions. In general, the electronic control unit includes a substrate on which electronic components are mounted. As an example of such a substrate, JP 2002-58129 A discloses a technology related to an electrical connection box including a wiring board having a plurality of reinforcing projections for supporting male terminals for fuse connections.

SUMMARY OF THE INVENTION

[0004] On a substrate such as a wiring board disclosed in JP 2002-58129 A, a configuration in which an electric wire is connected to a terminal having one end joined to the substrate can also be considered. When a plurality of electric wires are routed on a substrate, a routing board arranged in parallel with the substrate is further provided, a plurality of electric wires are routed on the routing board, and end portions of the electric wires may be connected to a part of a terminal exposed on the routing board in advance. However, when, for example, a plurality of terminals exposed on the routing board are randomly arranged, and a large number of electric wires are intermingled on a routing board, there is room for improvement in order to increase degrees of freedom for wiring.

[0005] An object of the present disclosure is to provide an electronic control unit inside of which degrees of freedom for wiring are increased.

[0006] An electronic control unit according to an aspect of an embodiment includes a substrate on which electronic components are mounted; a plurality of terminals having first ends connected to the substrate; a plurality of electric wires having a first end portion connected to one of second ends of the plurality of terminals on an opposite side of the first ends; and a routing board having a base in a shape of a plate, arranged in parallel with the substrate; in which the routing board routes the plurality of electric wires introduced from outside on a routing surface of the base opposite to a back surface facing the substrate; the second ends of at least one of the terminals are located between the base and the substrate; and the base has an opening partially facing the terminals having the second ends located between the base and the substrate and used for introducing the electric wires.

[0007] According to the configuration above, it is possible to provide an electronic control unit which increases degrees of freedom for wiring therein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of an electronic control unit according to one embodiment.

[0009] FIG. 2 is a perspective view of a circuit unit included in the electronic control unit.

[0010] FIG. 3 is an enlarged perspective view of electric wire connections corresponding to part III of FIG. 2.

[0011] FIG. 4 is a plan view of a circuit unit including an example of wiring on a routing board.

[0012] FIG. 5 is an enlarged plan view of the electric wire connections corresponding to part V in FIG. 4.

[0013] FIG. 6 is a side view of the electric wire connections corresponding to a cross section VI-VI in FIG. 5.

[0014] FIG. 7 is a cross-sectional view of the electric wire connections corresponding to a cross section VII-VII in FIG. 5.

[0015] FIG. 8 is a side view of the electric wire connections corresponding to a cross section VIII-VIII in FIG. 5.

[0016] FIG. 9 is a cross-sectional view of the electric wire connection corresponding to a cross section IX-IX in FIG. 5.

DETAILED DESCRIPTION OF THE INVENTION

[0017] Hereinafter, an electronic control unit according to one embodiment will be described in detail with reference to the drawings. Hereinafter, the term “electronic control unit” will be abbreviated as “ECU.” The dimensional ratios in the drawings are exaggerated for the sake of explanation and may differ from the actual ratios.

[0018] FIG. 1 is a perspective view of an ECU 1 according to one embodiment. FIG. 2 is a perspective view of a circuit unit 3 included in the ECU 1.

[0019] The ECU 1 is mounted in a vehicle such as an automobile, and controls various functions. The ECU 1 includes a substrate 10, a plurality of terminals 20, a plurality of electric wires 30, a connector 40, a routing board 50, a first cover 60, and a second cover 70.

[0020] Here, in the present embodiment, a unit consisting of the substrate 10, the plurality of terminals 20, a part of the plurality of electric wires 30, the routing board 50, the first cover 60, and the second cover 70 is denoted as “main body 2.” A unit including the substrate 10, the plurality of terminals 20, the plurality of electric wires 30, the connector 40, and the routing board 50 is denoted as “circuit unit 3.”

[0021] The substrate 10 is, for example, a printed circuit board (PCB) on which a plurality of electronic components 11 (see FIG. 6, for example.) are mounted in advance. Examples of the plurality of electronic components 11 include a CPU (Central Processing Unit) and a memory. In the present embodiment, the substrate 10 is regarded as a substantially rectangular flat plate which is parallel to a virtual XY plane defined by an X direction and a Y direction orthogonal to each other, and which has two ends in the X direction and two ends in the Y direction.

[0022] Each of the plurality of terminals 20 electrically connects one of the electric wires 30 to a predetermined circuit on the substrate 10. The terminals 20 are a flat, rod-shaped metal member. Each of the terminals 20 has a first end 20a which is one end, and a second end 20b which is the other end opposite to the first end 20a (see FIG. 6, for example). The first end 20a is connected to a predetermined circuit on the substrate 10. In the present embodiment, since the substrate 10 is regarded as a flat plate parallel to the XY plane, the terminals 20 are each arranged on a mounting surface 10a (see FIG. 6, for example) so as to project in the Z direction orthogonal to both the X and Y directions. The second end 20b connects a first end portion 31 (see FIG. 4) of one of the electric wires 30. The second end 20b includes, for example, two protruding pieces parallel to each other, and a core wire of the electric wires 30 is clamped between the two protruding pieces, so that the first end portion 31 can be easily connected.

[0023] In the present embodiment, the terminals 20 are roughly classified into two types, based on the height of the second end 20b when assembled in the circuit unit 3. Specifically, the second end 20b is located between a base 51 of the routing board 50 and the substrate 10 when assembled in the circuit unit 3 in at least one of the terminals 20 (hereinafter referred to as “first terminal”). On the other hand, at least a part of the second end 20b is exposed on the routing surface 51a of the base 51 when assembled in the circuit unit 3 in some of the terminals 20 (hereinafter referred to as a “second terminal”). Note that the terminals 20 illustrated in the drawings correspond to the first

terminal. The second terminal is not illustrated for convenience.

[0024] The electric wires **30** each electrically connect the connector **40** and one of the terminals **20**. The electric wires **30** are electric wires in which a core wire as a conductor, is coated with an insulator. Each of the electric wires **30** has the first end portion **31** which is one end, and a second end portion **32** which is the other end opposite to the first end portion **31**. As described above, the first end portion **31** is connected to one of the second ends **20b** of the terminals **20**. The second end portion **32** is connected to the connector **40**. In both of the first end portion **31** and the second end portion **32**, a part that actually comes into contact with the second ends **20b** or the connector **40** is a core wire portion partially exposed by removing the insulator.

[0025] The connector **40** is connected to an external connector (not illustrated) that is provided to an object to which the ECU **1** is attached. In the present embodiment, the ECU **1** includes, as an example, three connectors **40**, that is, a first connector **40a**, a second connector **40b**, and a third connector **40c**. The second end portion **32** of each of the electric wires **30** is connected to one of the first connector **40a**, the second connector **40b**, and the third connector **40c**. The connectors **40** are not directly provided on the substrate **10**, but are connected to the main body **2** of the ECU **1** only through the electric wire wires **30**.

[0026] The routing board **50** routes each of the electric wires **30** while connecting the first end portion **31** to the terminals **20** corresponding to each of the electric wires **30** inside the main body **2**. The routing board **50** is made of, for example, a synthetic resin, and is a flat member arranged in parallel with the substrate **10**. Since the routing board **50** is maintained in a stacked position with respect to the substrate **10**, it may also be called a "wiring layer." The routing board **50** has a base **51**, a side wall **52**, a pedestal **53**, at least one first terminal block **54** (see FIG. 6, for example.), a plurality of second terminal blocks **55**, and electric wire inlets **56**.

[0027] The base **51** is a plate facing the substrate **10** in the Z direction. The planar shape of the base **51** is equivalent to that of the substrate **10**. One surface of the base **51** is a routing surface **51a** on which a part of each of the electric wires **30** is wired. The other surface of the base **51** is on a side opposite to the routing surface **51a**, and is a back surface **51b** facing the substrate **10** (See FIG. 6, for example.).

[0028] The side wall **52** is continuous with an edge of the base **51**, and protrudes in a direction opposite to the side facing the substrate **10** in the Z direction so as to surround a routing space on the routing surface **51a**. However, the side wall **52** is not in the form of a frame surrounding entire four sides in the present embodiment, but is continuous with at least one of the electric wire inlets **56**.

[0029] The pedestal **53** is continuous with the edge of the base **51**, and protrudes in the Z direction to be at least partly in contact with an outer edge of the mounting surface **10a** of the substrate **10**. Thus, a constant distance between the base **51** and the substrate **10** is maintained.

[0030] The first terminal blocks **54** and the second terminal blocks **55** support the terminals **20** with the second end **20b** thereof being exposed. The arrangement of the terminals **20** depends on the circuit configuration of the mounting surface **10a** of the substrate **10**. Therefore, the arrangement of the first terminal blocks **54** and the second terminal blocks **55** is also determined based on the circuit configuration of the substrate **10**.

[0031] FIG. 3 is an enlarged perspective view of electric wire connections **4** corresponding to part III of FIG. 2. Here, the electric wire connections **4** refer to components in which the electric wires **30** are connected to the second end **20b** of the terminals **20** in the first terminal blocks **54** or the second terminal blocks **55**.

[0032] The first terminal blocks **54** support the second end **20b** of the first terminal of the plurality of terminals **20**. In the present embodiment, as an example, it is assumed that three terminals **20** are present as the first terminals, and in addition, three first terminal blocks **54** are present to individually support these terminals **20**. The first terminal blocks **54** have an opening groove **81** which exposes a part of the second end **20b** therein. For each of the electric wires **30**, a part of the

first end portion **31** is inserted into the opening groove **81** that is predetermined, so that the core wire of the first end portion **31** is connected to the second end **20b**. FIG. **3** illustrates a part of the opening grooves **81**, and openings **81a** of the opening grooves **81**, exposed from the routing surface **51a**. Hereinafter, the three electric wire connections **4** including the first terminal blocks **54** are referred to as a first electric wire connection **4a**, a second electric wire connection **4b**, and a third electric wire connection **4c**. The first electric wire connection **4a**, the second electric wire connection **4b**, and the third electric wire connection **4c** are lined up in a direction opposite to the X direction. The specific configuration of the electric wire connections **4** including the first terminal blocks **54** will be described in detail below.

[0033] The second terminal blocks **55** support a second end of a second terminal (not illustrated) among the terminals **20**. As will be described later, the first terminal blocks **54** are provided on the back surface **51b** of the base **51**, while the second terminal blocks **55** are provided on the routing surface **51a** of the base **51**. The basic shape of the second terminal blocks **55** itself is the same as that of the first terminal blocks **54**. Specifically, the second terminal blocks **55** supporting the second end of one of the terminals **20** have a shape in which the first terminal blocks **54** are inverted in the Z direction with respect to the base **51**. The second terminal blocks **55** may have a shape in which a plurality of second terminal blocks **55** supporting the second end of one of the terminals **20** are integrated in parallel and continuously.

[0034] In the present embodiment, three first terminals are present, and terminals among the plurality of terminals **20** other than first terminals are second terminals. However, among the plurality of terminals **20**, the number of the first terminals may be larger than the number of the second terminals. Furthermore, all of the plurality of terminals **20** may be first terminals. That is, the presence of the second terminals is not essential. Therefore, when all of the plurality of terminals **20** are the first terminals, all of the plurality of terminal blocks present in the routing board **50** become the first terminal blocks **54**.

[0035] The electric wire inlets **56** introduce the electric wires **30** onto the routing surface **51a** of the base **51** from the outside of the routing board **50** while supporting a part of the electric wires **30**. At least one of the electric wire inlets **56** is provided as a convex portion at an outer edge of the routing board **50** and on the routing surface **51a**. In the present embodiment, as an example, three electric wire inlets **56** are present: a first electric wire inlet **56a** extending in the Y direction, and a second electric wire inlet **56b** and a third electric wire inlet **56c**, each extending in the X direction. When routing the electric wires **30**, not necessarily all of the electric wire inlets **56** are used. FIG. **1** illustrates, as an example, a configuration in which all the electric wires **30** are pulled into the body **2** from the connector **40** only through the first electric wire inlet **56a**.

[0036] The first cover **60** and the second cover **70** are combined with each other to accommodate at least the substrate **10**, the plurality of terminals **20**, and the routing board **50** included in the circuit unit **3**. The first cover **60** and the second cover **70** may be made of a synthetic resin or metal, for example.

[0037] The first cover **60** is a cover to be assembled from a side facing the substrate **10**. The first cover **60** has a first flat plate **61** (see FIG. **6**) covering the substrate **10** and a first frame **62** continuous with an edge of the first flat plate **61**. The first frame **62** projects in the Z direction in which the first frame **62** is assembled with respect to the second cover **70**, so as to surround the substrate **10**, for example. A part of a first tip edge **62a** of the first frame **62** is close to the electric wire inlets **56** of the routing board **50** in a non-contact manner.

[0038] The second cover **70** is a cover to be assembled from a side facing the routing surface **51a** of the routing board **50**. The second cover **70** has a second flat plate **71** (see FIG. **6**) covering the routing board **50**, and a second frame **72** continuous with an edge of the second flat plate **71**. The second frame **72** projects in a direction opposite to the Z direction, which is the direction to be assembled with respect to the first cover **60**, so as to surround at least a part of the routing board **50**, for example. A part of a second tip edge **72a** of the second frame **72** is close to the electric wire

inlet 56 of the routing board 50 in a non-contact manner. A part of an area between the first tip edge 62a of the first frame 62 of the first cover 60 and the second tip edge 72a of the second frame portion 72 of the second cover 70 is an open area through which the plurality of electric wires 30 pass inside and outside the main body 2.

[0039] The first cover 60 has a plurality of first locking portions 63 in the first frame 62. On the other hand, the second cover 70 has a plurality of second locking portions 73 in the second frame 72. The second cover 70 is combined with the first cover 60 by engaging the second locking portions 73 with the first locking portions 63.

[0040] Next, a specific configuration of the electric wire connections 4 including the first terminal blocks 54 will be described.

[0041] FIG. 4 is a plan view of the circuit unit 3 including an example of wiring on the routing board 50. FIG. 5 is an enlarged plan view of the electric wire connections 4 including the first terminal blocks 54 corresponding to part V in FIG. 4.

[0042] Here, although a large number of the electric wires 30 can actually be wired in the routing board 50, only two electric wires 30, the first electric wire 30a and the second electric wire 30b, will be exemplified below for the sake of simplicity.

[0043] The first electric wire 30a is an example of the electric wires 30 pulled into the main body 2 from a connector (not illustrated) similar to the connector 40 through the second electric wire inlet 56b. The first electric wire 30a is linearly wired from the second electric wire inlet 56b toward the first electric wire connection 4a on the routing surface 51a, for example, passing through gaps in the plurality of second terminal blocks 55. The first end portion 31 of the first electric wire 30a is connected to the second end 20b of the terminals 20 located in the first electric wire connection 4a.

[0044] The second electric wire 30b is an example of the electric wires 30 pulled into the main body 2 from the connector 40 through the first electric wire inlet 56a. The second electric wire 30b is linearly routed from the first electric wire inlet 56a to the second end 20b of the terminals 20 supported by one of the second terminal blocks 55, for example, passing through gaps in the plurality of second terminal blocks 55 on the routing surface 51a. Hereinafter, the second terminal block 55 supporting the second end 20b to which the first end portion 31 of the second electric wire 30b is connected is referred to as a “connection terminal block 55a” for convenience. The connection terminal block 55a is provided in the vicinity of another side wall 52 opposed in the X direction to the side wall 52 in which the first electric wire inlet 56a is provided.

[0045] The second electric wire 30b passes through an upper portion of the openings 81a corresponding to each of the first electric wire connection 4a, the second electric wire connection 4b, and the third electric wire connection 4c. The second electric wire 30b passes above the first electric wire 30a connected to the first electric wire connection 4a. Here, “above” with respect to each member refers to an area corresponding to an upper side when the routing surface 51a is a horizontal plane, and the normal direction of the routing surface 51a aligned with the Z direction is from bottom to top.

[0046] FIGS. 6 and 7 are views of the first electric wire connection 4a, the second electric wire connection 4b, and the third electric wire connection 4c as viewed in the Y direction. FIG. 6 is a side view of the electric wire connections 4 corresponding to a cross section VI-VI in FIG. 5. FIG. 7 is a cross-sectional view of the electric wire connections 4 corresponding to a cross section VII-VII in FIG. 5. The cross section in FIG. 7 is in an XZ plane passing through an intermediate position in a thickness direction corresponding to the Y direction of the terminals 20 supported by the first terminal blocks 54.

[0047] FIGS. 8 and 9 are views of the first electric wire connection 4a viewed in the negative X direction. FIG. 8 is a side view of the first electric wire connection 4a corresponding to a cross section VIII-VIII in FIG. 5. FIG. 9 is a cross-sectional view of the first electric wire connection 4a corresponding to a cross section IX-IX in FIG. 5. The cross section in FIG. 9 is in the YZ plane passing through an intermediate position in a width direction corresponding to the X direction of

the terminals **20** supported by the first terminal blocks **54**.

[0048] The three electric wire connections **4** including the first terminal blocks **54** have the same configuration. Therefore, the first electric wire connection **4a** will be mainly described with respect to the shape or the like.

[0049] First, the second end **20b** of the terminals **20** as the first terminals connects the electric wires **30** by clamping a core wire of the electric wires **30** between the two protruding pieces as described above. Hereinafter, the position in which the second end **20b** clamps and connects the first electric wire **30a** is defined as a “connection point 5,” as illustrated in FIGS. 7 and 9. The second end **20b** is located between the base **51** and the substrate **10**, so that the connection point 5 is located close to the substrate **10** relative to the position of the base **51**.

[0050] The first terminal blocks **54** are provided for each of the terminals **20** in the present embodiment, and therefore, three first terminal blocks **54** are present for the first electric wire connection **4a**, the second electric wire connection **4b**, and the third wire connection **4c**. As in the example of the second terminal blocks **55** described above, the first terminal blocks **54** may have a shape such that the three first terminal blocks **54** supporting the second end **20b** of each of the terminals **20** are integrated in parallel and continuously.

[0051] The overall schematic shape of the first terminal blocks **54** is a cuboid with one end continuous with the back surface **51b** of the base **51**. The first terminal blocks **54** have a terminal through-hole **80** and the opening groove **81**.

[0052] The terminal through-hole **80** accommodates the second end **20b** of each of the terminals **20**. The terminal through-hole **80** is penetrating in the Z direction, a direction in which the base **51** faces the substrate **10**. One opening of the terminal through-hole **80** is located at a tip of the first terminal block **54** facing the substrate **10**. The other opening of the terminal through-hole **80** is located at a bottom surface **81b** of the opening groove **81**. However, a part of a wall defining the terminal through-hole **80** may be a pair of grooves extending from the bottom surface **81b** to the opening **81a** in the Z direction and exposed from the routing surface **51a**. In other words, if exposed ends of the grooves are also regarded as a part of the opening **81a**, the part of the opening **81a** can be regarded as facing each of the terminals **20** in the Z direction. In the present embodiment, the direction in which the second end **20b** clamps the first electric wire **30a** (Hereinafter referred to as a “clamping direction.”) is in the X direction, so that the pair of grooves in which a part of the wall defining the terminal through-hole **80** is exposed from the routing surface **51a** also face each other in the X direction.

[0053] The opening groove **81** has at least the opening **81a** formed as a part of the base **51** on the routing surface **51a**, and a bottom surface **81b** separated from the opening **81a** toward the substrate **10**, allowing the first end portion **31** of the first electric wire **30a** to be introduced from the opening **81a**. The opening groove **81** communicates with the terminal through-hole **80**, and exposes a part of the second end **20b** of the terminals **20** therein. In order to satisfy conditions that the first end portion **31** is introduced from the opening **81a**, and the second end **20b** clamps and connects the introduced first end portion **31**, a first width **W1** of the opening **81a** in a direction along a clamping direction is set larger than a dimension of an outer diameter **D** of the first electric wire **30a**. On the other hand, a second width **W2** in a direction orthogonal to the first width **W1** of the opening **81a** is set larger than the first width **W1** to the extent that a tip of the first end portion **31** can be exposed on the routing surface **51a** as in the present embodiment.

[0054] The bottom surface **81b** of the opening groove **81** is substantially parallel to the routing surface **51a**, and is set at a position at a distance **L** from the back surface **51b** of the base **51**, for example. The distance **L** may be set in advance so that the connection point 5 is set at a desired position when the first end portion **31** of the first electric wire **30a** introduced into the opening groove **81** is in contact with the bottom surface **81b**. It should be noted that the opening groove **81** may be opened from the back surface **51b** to the bottom surface **81b** of the base **51** so as to be opened to a spatial region between the base **51** and the substrate **10**.

[0055] Next, the effect of the ECU **1** will be described.

[0056] The ECU **1** includes the substrate **10** on which the electronic components **11** are mounted, the plurality of terminals **20** having the first ends **20a** connected to the substrate **10**, and the plurality of electric wires **30** having the first end portion **31** connected to the second end **20b** of the terminals **20** on an opposite side of the first ends **20a**. The ECU **1** also includes the routing board **50** having the base **51** in a shape of a plate, arranged in parallel with the substrate **10**. The routing board **50** routes the plurality of electric wires **30** introduced from the outside on the routing surface **51a** of the base **51** opposite to the back surface **51b** facing the substrate **10**. The second end **20b** of at least one of the terminals **20** is located between the base **51** and the substrate **10**. The base **51** has an opening **81a** partially facing the terminals **20** having the second end **20b** located between the base **51** and the substrate **10** and introducing the electric wires **30**.

[0057] Here, for comparison, it is assumed that a virtual terminal block **154** equivalent to the second terminal block **55** is provided at the position of the first electric wire connection **4a** instead of the first terminal block **54** in FIGS. **6** and **8**. In other words, the virtual terminal block **154** supports a second terminal at least partially exposed on the routing surface **51a** of the base **51**. The height of a tip face **154a** in the Z direction of the virtual terminal block **154** can be set equal to the position of a tip edge **52a** of the side wall **52** of the routing board **50**.

[0058] According to the comparative example in which the virtual terminal block **154** is present, the first electric wire **30a** can be wired in a wiring path equivalent to the above example in which the virtual terminal block **154** is not present, by connecting the first end portion **31** to the second end **20b** supported by the virtual terminal block **154**. However, the second electric wire **30b** cannot pass through an upper portion of the opening **81a** included in the first electric wire connection **4a** due to the virtual terminal block **154** present at the position of the first electric wire connection **4a**. Therefore, as illustrated in FIG. **4**, a virtual electric wire **130b** in place of the second electric wire **30b** is routed on a wiring path that avoids not only the virtual terminal block **154** but also a plurality of the second terminal blocks **55**. In other words, the wiring path of the virtual electric wire **130b** is assumed to be more complicated than that of the second electric wire **30b**. Moreover, the length of the virtual electric wire **130b** is assumed to be longer than that of the second electric wire **30b**.

[0059] In the ECU **1** according to the present embodiment, for some of the terminals **20**, the second end **20b** is shaped to be located between the base **51** and the substrate **10**, so that the position of the connection point **5** connecting to the first end portion of the desired one of the electric wires **30** is not set on the routing surface **51a**. Therefore, according to the ECU **1**, when some of the electric wires **30** such as the second electric wire **30b** should be wired on a simple linear path, for example, the second end **20b** of the terminals **20** can be prevented from being arranged at a position crossing the wiring path. In addition, although the wiring path is set on the routing surface **51a**, the fact that the second end **20b** is prevented from being arranged at a position crossing the wiring path means that the second terminal block **55** need not be arranged on the routing surface **51a**. Therefore, as illustrated in FIGS. **4** and **5**, the second electric wire **30b** is less likely to interfere with the electric wires **30** such as the first electric wire **30a** connected to one of the terminals **20**, or with a terminal block on the routing surface **51a** such as the second terminal block **55**, degrees of freedom for wiring the electric wires **30** can be increased. In particular, instead of the examples illustrated in the drawings, degrees of freedom for wiring the electric wires **30** can be further improved if the second end **20b** is arranged between the base **51** and the substrate **10** in many of the terminals **20**.

[0060] As described above, according to the present embodiment, it is possible to provide the ECU **1** which increases the degrees of freedom for wiring therein.

[0061] According to the ECU **1**, the wiring path is simplified for each of the electric wires **30** by increasing the degrees of freedom for wiring, wire saving of the plurality of electric wires **30** is possible, and the main body **2** can be miniaturized or reduced in weight by increasing the wiring space on the routing board **50**.

[0062] In addition, in the ECU **1**, the routing board **50** may have a plurality of terminal blocks which support the second end **20b** with a part thereof exposed. The terminal blocks which support the second end **20b** located between the base **51** and the substrate **10** may be provided on the back surface **51b** of the base **51**.

[0063] In the above example, the terminal block which supports the second end **20b** located between the base **51** and the substrate **10** is the first terminal block **54** among the plurality of terminal blocks. On the other hand, the terminal blocks other than the first terminal blocks **54** are the second terminal blocks **55**.

[0064] According to the ECU **1**, the terminals **20** having the second end **20b** located between the base **51** and the substrate **10** are also supported by the first terminal blocks **54**, so that the connection stability between the terminals **20** and the electric wires **30** can be prevented from deteriorating.

[0065] Furthermore, the ECU **1** may include the connectors **40** to which the second end portion **32** opposite to the first end portion **31** of each of the electric wires **30** is connected.

[0066] In this ECU **1**, the connectors **40** are only connected to the main body **2** via the plurality of electric wires **30**, so that positions of the connectors **40** can be freely changed relative to the main body **2**. Therefore, according to the ECU **1**, it is possible to avoid, for example, the occurrence of troubles such as reduced space in a passenger compartment due to bulging of external electric wires around the main body **2**.

[0067] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. An electronic control unit comprising: a substrate on which electronic components are mounted; a plurality of terminals having a first end connected to the substrate; a plurality of electric wires having a first end portion connected to one of the second ends of the plurality of terminals on an opposite side to the first end; and a routing board having a base in a shape of a plate, arranged in parallel with the substrate; wherein: the routing board routes the plurality of electric wires introduced from outside on a routing surface of the base opposite to a back surface facing the substrate; the second ends of at least one of the terminals are located between the base and the substrate; and the base has an opening partially facing the terminals having the second ends located between the base and the substrate and introducing the electric wires.
 2. The electronic control unit according to claim 1, wherein the routing board has a plurality of terminal blocks which support the second ends with a part thereof exposed; and the terminal blocks which support the second ends located between the base and the substrate is provided on the back surface of the base.
 3. The electronic control unit according to claim 1, further comprising connectors to which a second end portion opposite to the first end portion of each of the electric wires is connected.
-